

[CSE322] TOC-2025: Quiz 3 Rubric

Full Marks: 20, Time: 40 minutes

(1) Given two sets X and Y , $X\Delta Y = (X \cup Y) - (X \cap Y)$. Let B and D be two regular languages over a nonempty finite alphabet Σ . Furthermore, assume that $M_B = (Q_B, \Sigma, \delta_B, q_B, F_B)$ and $M_D = (Q_D, \Sigma, \delta_D, q_D, F_D)$ be the DFAs such that $L(M_B) = B$ and $L(M_D) = D$.

Construct the DFA $M = (Q, \Sigma, \delta, q_0, F)$ as follows.

- $Q = Q_B \times Q_D$.
- $q_0 = (q_B, q_D)$.
- $\delta((p_B, p_D), a) = (\delta_B(p_B, a), \delta_D(p_D, a))$.
- $F = \{(p_1, p_2) \mid (p_1 \in F_B \text{ and } p_2 \notin F_D) \text{ or } (p_1 \notin F_B \text{ and } p_2 \in F_D)\}$.

Prove or disprove that $L(M) = B\Delta D$.

(7 Marks)

Solution: [Proof]

Proof: Key observation - $\forall w \in \Sigma^*, \hat{\delta}((p_B, p_D), w) = (\hat{\delta}_B(p_B, w), \hat{\delta}_D(p_D, w))$

Forward direction $[L(M) \subseteq B\Delta D]$

Let $w \in L(M)$. Therefore, $\hat{\delta}((q_B, q_D), w) \in F \implies (\hat{\delta}_B(q_B, w), \hat{\delta}_D(q_D, w)) \in F$

Therefore, via the definition of F , we get

$$(\hat{\delta}_B(q_B, w) \in F_B \text{ and } \hat{\delta}_D(q_D, w) \notin F_D) \text{ or } (\hat{\delta}_B(q_B, w) \notin F_B \text{ and } \hat{\delta}_D(q_D, w) \in F_D)$$

$$\equiv (w \in L(B) \text{ and } w \notin L(D)) \text{ or } (w \notin L(B) \text{ and } w \in L(D))$$

$$\equiv (w \in B \text{ and } w \notin D) \text{ or } (w \notin B \text{ and } w \in D) \equiv w \in (B \cap D^C) \cup (B^C \cap D) \equiv w \in B\Delta D$$

Therefore, we get $L(M) \subseteq B\Delta D$.

Backward direction $[B\Delta D \subseteq L(M)]$

Let $w \in B\Delta D$. Therefore, $w \in (B \cup D) - (B \cap D) = (B \cup D) \cap (B^C \cup D^C)$. Thus, $w \in B \cup D$ and $w \in B^C \cup D^C$. This is equivalent to exactly one of the following two mutually exclusive cases -

Case 1 - $w \in B$ and $w \notin D$

Since $w \in B$ and $B = L(M_B)$, we get $\hat{\delta}_B(q_B, w) \in F_B$. Similarly, since $w \notin D$ and $D = L(M_D)$, we get $\hat{\delta}_D(q_D, w) \notin F_D$.

Now, examining $\hat{\delta}((q_B, q_D), w) = (\hat{\delta}_B(q_B, w), \hat{\delta}_D(q_D, w))$. Since the first element of the tuple is in F_B and the second is not in F_D , we get (through the definition of set F) that $\hat{\delta}((q_B, q_D), w) \in F$. Therefore, $w \in L(M)$

Case 2 - $w \in D$ and $w \notin B$

Since $w \in D$ and $D = L(M_D)$, we get $\hat{\delta}_D(q_D, w) \in F_D$. Similarly, since $w \notin B$ and $B = L(M_B)$, we get $\hat{\delta}_B(q_B, w) \notin F_B$.

Now, examining $\hat{\delta}((q_B, q_D), w) = (\hat{\delta}_B(q_B, w), \hat{\delta}_D(q_D, w))$. Since the first element of the tuple is not in F_B and the second is in F_D , we get (through the definition of set F) that $\hat{\delta}((q_B, q_D), w) \in F$. Therefore, $w \in L(M)$

Since both cases are exhaustive, we can conclude that $w \in L(M)$. Therefore, $B\Delta D \subseteq L(M)$. $\square$

(2) Let B be any language over a nonempty finite alphabet Σ . For every $k \geq 1$, define $B^k = \{w_1w_2 \dots w_k \mid \text{for all } 1 \leq i \leq k, w_i \in B\}$. Subsequently, define

$$B^+ = B \cup B^2 \cup \dots$$

Prove that $B = B^+$ if and only if $B^2 \subseteq B$.

(8 Marks)

Solution: [Proof]

Forward direction [$\implies$]

To prove : $B = B^+ \implies B^2 \subseteq B$.

Proof: Since $B^+ = B \cup B^2 \cup \dots = \bigcup_{i=1}^{\infty} B^i \implies B^2 \subseteq B^+$

But $B^+ = B$. Therefore, $B^2 \subseteq B$, proving our claim. $\square$

Backward direction [$\impliedby$]

To prove : $B^2 \subseteq B \implies B = B^+$.

Proof: [Proof via contradiction]

Let us assume that $B^2 \subseteq B$ and $B \neq B^+$. Since $B \subseteq (B \cup B^2 \cup \dots) = B^+$, we must have $B^+ \not\subseteq B$. Therefore, $B^+ \setminus B \neq \emptyset$. ($\setminus$ is the set minus operation, $A \setminus B = A \cap B^C$)

Choose such a string $x \in B^+ \setminus B$ for which the number of factors in a decomposition

$$x = w_1w_2 \cdots w_k \quad \text{with } w_i \in B$$

is minimum. We see, $k \geq 2$ as $k = 1$ would put x in B .

Write $x = yz$, where

$$y = w_1w_2 \cdots w_{k-1} \in B^{k-1} \quad \text{and} \quad z = w_k \in B.$$

By the minimality of k , $y \notin B^+ \setminus B$. Otherwise, y would admit a shorter decomposition than x , contradicting the minimality of k . Since $y \in B^{k-1} \subseteq B^+$, it follows that $y \in B$.

Hence, we have $y \in B$ and $z \in B$, which implies

$$yz \in B^2.$$

Since $B^2 \subseteq B$ by assumption, we conclude that

$$x = yz \in B,$$

which contradicts the choice of $x \notin B$.

Therefore, our assumption that $B^2 \subseteq B$ and $B \neq B^+$ is false. Hence, $B = B^+$.

□

Therefore, by proving both directions of the implication we have proved

$$B = B^+ \iff B^2 \subseteq B \quad \blacksquare$$

Alternative proof of backward direction [Induction]

Given that $B^2 \subseteq B$, we need to show $B = B^+$.

Claim - $B \subseteq B^+$

Trivially true as $B \subseteq B \cup B^2 \cup \dots = B^+$

Claim - $B^+ \subseteq B$

We prove this by employing the argument of induction. We aim to show that $\forall k \in \mathbb{N}^+, B^k \subseteq B$. For this we **perform induction on k** .

Base Case - $k = 1$. We can see that $B^k = B^1 = B \subseteq B$. Therefore the statement holds for the base case.

Induction Hypothesis - The induction hypothesis here is

$$\forall n \in \{1, 2, \dots, k\}, B^n \subseteq B$$

Induction step - We therefore must prove that $B^{k+1} \subseteq B$. Let $x \in B^{k+1}$.

Therefore, we can split x into $x_1x_2\dots x_kx_{k+1}$ such that $x_1, x_2, \dots, x_k, x_{k+1} \in B$. The key idea here is to see that the string $x_1x_2\dots x_k \in B^k$ (Another way to mention this is $B^{k+1} = \{xy : x \in B^k, y \in B\}$).

Since $B^k \subseteq B$ from the induction hypothesis, then the string $x_1x_2\dots x_k \in B$. Therefore, our string $x = x_1x_2\dots x_kx_{k+1}$ can be split into two strings, both in B as $x_1x_2\dots x_k \in B$ and $x_{k+1} \in B$. Therefore, $x \in B^2$.

However, since $B^2 \subseteq B$, we get $x \in B$. Since x was an arbitrary string in B^{k+1} , we get that $B^{k+1} \subseteq B$.

Thus using the induction argument we get that $\forall k \in \mathbb{N}^+, B^k \subseteq B$. Therefore, we get

$$B^+ = (B \cup B^2 \cup B^3 \cup \dots) \subseteq (B \cup B \cup B \cup \dots) = B$$

Therefore, we get $B^+ \subseteq B$.

Through both claims, we get that $B \subseteq B^+$ and $B^+ \subseteq B \implies B = B^+$ ■